

Coffee Shop Live Analysis v2 — Transaction Analysis

Business Analytics Report

Project: Coffee Shop Live Analysis v2

Dataset: coffee_shop_transactions.csv

Analysis date: 2026-09-25

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Generated by the AI Data Scientist Platform

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1. Executive Summary

This report analyzes **coffee_shop_transactions.csv** (69,785 rows, 12 columns) covering 2025-01-01 to 2099-01-01.

Business question: *How is the business performing (revenue, refunds), who are the best customers, which customers are likely to return, and what should revenue look like in the coming months?*

Key verified findings

- The original dataset contained 6,523 missing value(s) and 45 duplicate row(s), which have since been cleaned.
- Total revenue: **61,646,516**, refunds: 39,508 (0.06% of transactions), average order value 884.
- 705 of 1150 identified customers are predicted to return within 90 days.
- Revenue forecast for the next 3 month(s): 2026-04: 8,527,531, 2026-05: 7,277,674, 2026-06: 7,682,696.

Top recommendations

- 'Hot Coffee' is the top revenue category at 13,077,917 — consider prioritizing inventory, staffing, and promotions around it.
- Peak revenue hour is 22:00 (5,590,365) — align staffing levels and time-limited promotions with this window.
- Refund rate is 0.06% of transactions — low, with no sign of a widespread product or service-quality issue.
- 230 customers (20.0% of identified customers) are in the 'At Risk' or 'Lost' RFM segments, representing 5,457,134 in historical spend — a targeted re-engagement offer to this group could recover meaningful revenue.
- Predicted return likelihood is 74.2% for customers who have redeemed loyalty points vs. 56.4% for those who haven't — loyalty-program members show a 17.8 point higher predicted return rate, suggesting the loyalty program is genuinely associated with retention.

2. About the Dataset

coffee_shop_transactions.csv contains 69,785 rows and 12 columns. It spans the period 2025-01-01 to 2099-01-01 (based on Transaction Time).

Column dictionary

Column	Type	Missing	Notes
Transaction Id	Identifier	0.0%	
Customer Id	Identifier	0.0%	most common: Walk-in / Unknown
Transaction Time	Date/Time	0.0%	
Branch	Category	0.0%	most common: Karachi - DHA
Item	Category	0.0%	most common: Blueberry Muffin
Category	Category	0.0%	most common: Hot Coffee
Quantity	Numeric	0.0%	typical range -3.00–3.00
Unit Price Pkr	Numeric	0.0%	typical range 420.00–1,026.00
Discount Pkr	Numeric	0.0%	typical range 0.00–308.00
Payment Method	Category	0.0%	most common: Card
Loyalty Points Redeemed	Numeric	0.0%	typical range 0.00–299.00
Total Pkr	Numeric	0.0%	typical range -2,640.00–3,072.00

3. Data Quality & Cleaning

Original dataset quality (as uploaded)

Total rows	Missing cells	Duplicate rows	Issues detected
69,830	6,523	45	7

Duplicate Rows

Column	Finding	Affected
(whole dataset)	45 duplicate row(s) detected.	45

Missing Values

Column	Finding	Affected
customer_id	'customer_id' has 4429 missing value(s) (6.3%).	4429
payment_method	'payment_method' has 2094 missing value(s) (3.0%).	2094

Invalid Dates

Column	Finding	Affected
transaction_time	'transaction_time' has 6 value(s) that are unparseable, in the future, or far outside the rest of the column's date range.	6

Impossible Values

Column	Finding	Affected
quantity	'quantity' has 44 negative value(s) that are rare exceptions among otherwise non-negative values — likely a data-entry error, not real signal.	44
total_pkr	'total_pkr' has 44 negative value(s) that are rare exceptions among otherwise non-negative values — likely a data-entry error, not real signal.	44

Outliers

Column	Finding	Affected
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total_pkr	'total_pkr' has 8270 statistical outlier(s) outside [87.5, 1467.5].	8270
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Cleaning performed

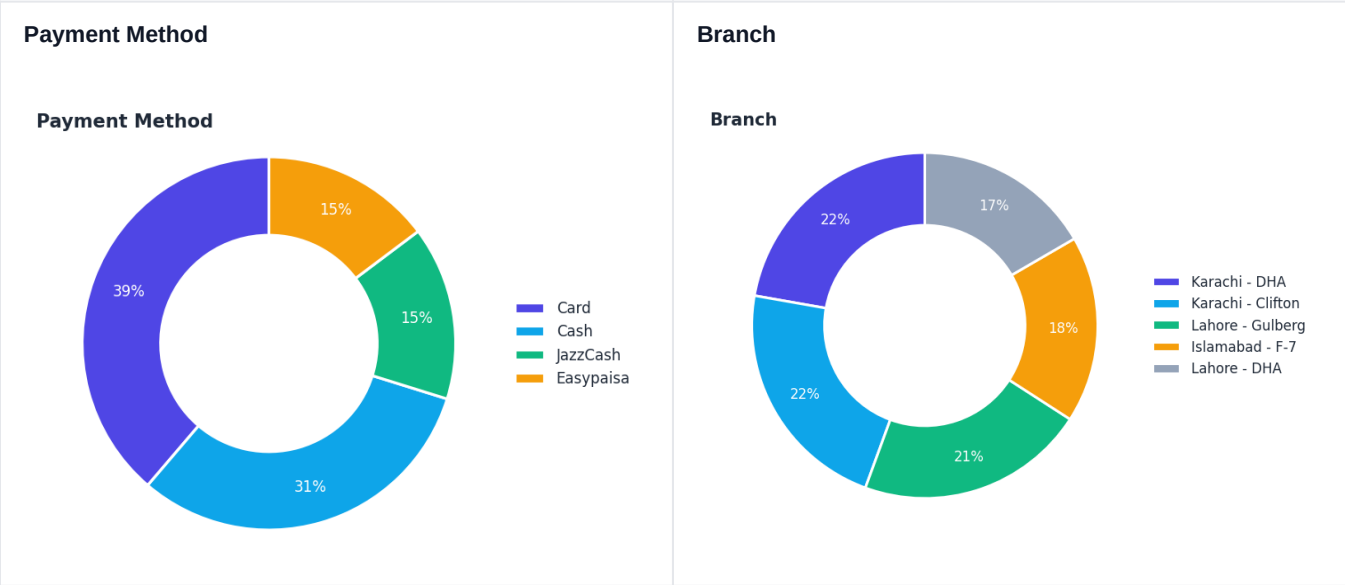
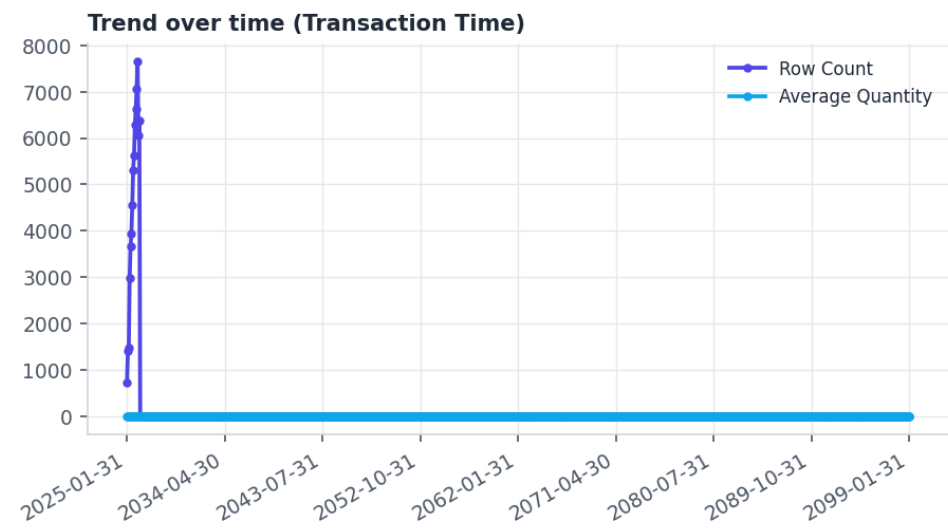
Problem	Before	Action	After	Reason
Duplicate rows	45 duplicate row(s)	Removed duplicate rows	0 duplicate rows	Duplicate records would otherwise double-count the same transaction/customer.
Missing Customer Id	4429 missing	Filled using 'label_unknown'	0 missing	Selected automatically based on the column's data type.
Missing Payment Method	2094 missing	Filled using the most frequent value (mode)	0 missing	The most representative value for a categorical field.
Text-formatted numbers in Unit Price Pkr	stored as text with units (e.g. '4.2 km')	Converted to a clean numeric value	numeric	100% of values matched a number-with-unit pattern.
Inconsistent capitalization in Branch	5 variant label(s) (e.g. 'rainy' vs 'Rainy')	Merged into a single canonical label	consistent labels	Values differing only by letter case would otherwise be treated as different categories.

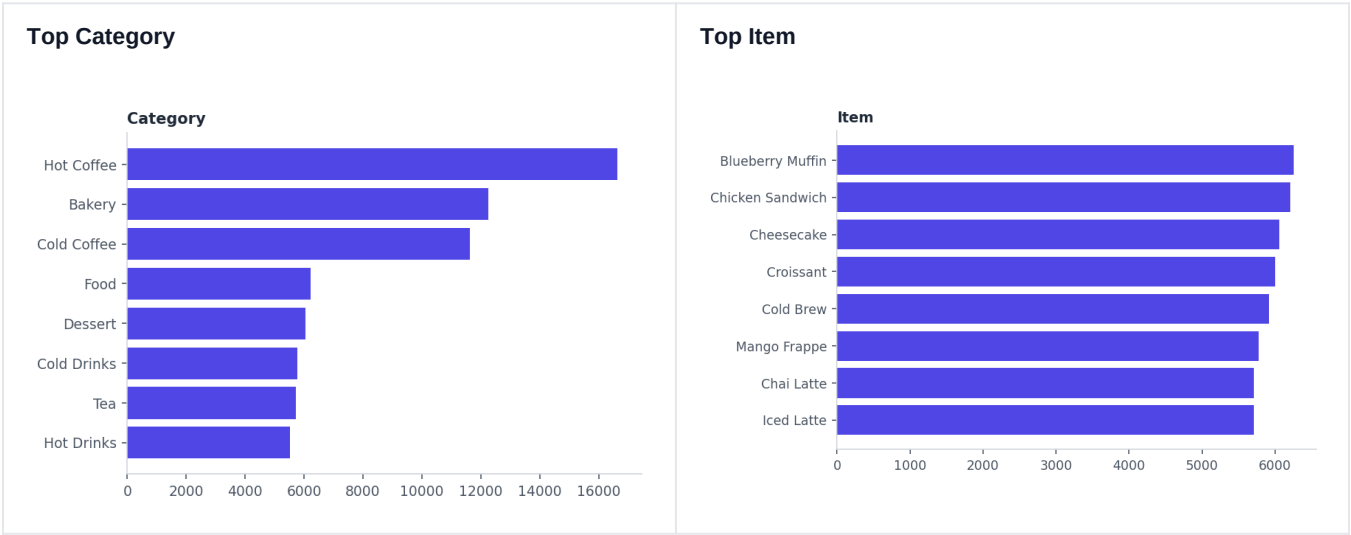
After cleaning: 0 missing cell(s), 0 duplicate row(s), 69,785 rows remain. The original file itself was never modified — cleaning produced a separate working copy.

4. Key EDA Findings

Total Rows	Average Unit Price Pkr	Total Quantity	Total Discount Pkr
69,785	716	87,015	679,961

Trend over time (Transaction Time)





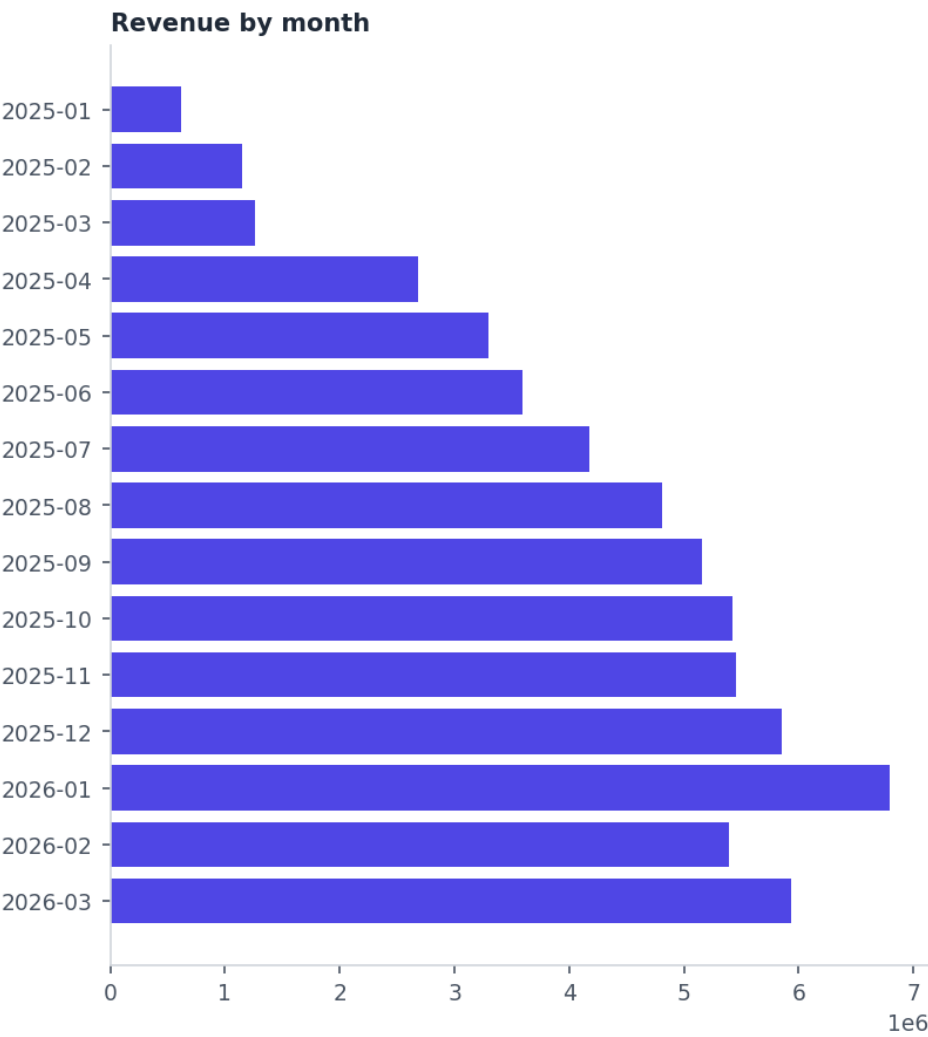
Quantity breakdown

Average: 1.25 (range -3.0–3.0)

Value	Count	Percent
3	3473	5.0%
2	10381	14.9%
1	55887	80.1%
-1	38	0.1%
-2	3	0.0%
-3	3	0.0%

5. Revenue Analytics

Total revenue	Refunds	Refund rate	Avg order value
61,646,516	39,508	0.06%	884



Revenue by category		Revenue by branch	
Category	Revenue	Branch	Revenue
Hot Coffee	13,077,917	Karachi - DHA	13,736,997
Cold Coffee	11,651,863	Karachi - Clifton	13,688,441
Food	7,437,353	Lahore - Gulberg	13,196,294
Bakery	7,021,846	Islamabad - F-7	10,733,957
Dessert	6,713,461	Lahore - DHA	10,251,319
Cold Drinks	6,367,989		
Hot Drinks	5,274,543		
Tea	4,062,036		

6. Customer Analytics & RFM Segments

1150 identified customer(s) analyzed (walk-in/unidentified transactions excluded).

Segment	Customers	% of customers	Avg. monetary	Avg. frequency	Avg. recency (days)
Loyal Customers	130	11.3%	145,509	105.1	19.5
Needs Attention	668	58.1%	26,712	19.3	24.8
Champions	90	7.8%	170,357	123.9	0.3
At Risk	159	13.8%	32,576	23.3	227.7
Lost	71	6.2%	3,910	2.7	262.9
New Customers	32	2.8%	3,765	2.7	0.4

7. Customer Return Prediction

Of **1150** identified customers, **705** are predicted **likely to return** within the next 90 days, **404** are **at risk of not returning**, and **41** are **uncertain** ($705 + 404 + 41 = 1150$).

Trained and validated on a strictly time-based split (features from history before a past cutoff, labels from real outcomes after it) — holdout accuracy: 92.9%.

Risk group	Predicted-probability band	Customers
Likely to return	$\geq 60\%$ predicted probability	705
At risk of not returning	$< 40\%$ predicted probability	404
Uncertain	40%-60% predicted probability	41

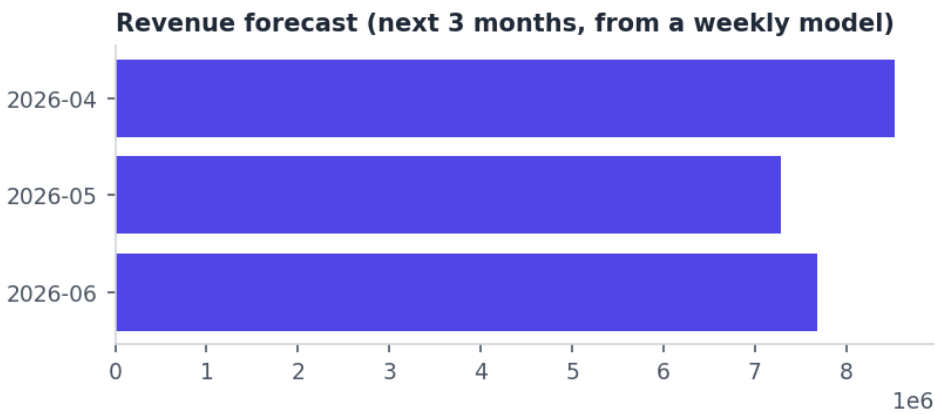
8. Sales Forecast

Method: **Holt's linear trend**, selected from 4 candidate method(s) by rolling backtest on 65 week(s) of history — modeled weekly, shown here aggregated to monthly.

Selected by rolling backtest (lowest out-of-sample error across 7 historical fold(s)) — never fit to any held-out answer key. Accounts for the historical weekly revenue trend.

Method comparison (rolling backtest — never the answer to this forecast)

Method	Backtest MAPE	Folds
Holt's linear trend selected	7.38%	7
Naive (repeat last week)	12.37%	7
Regression with calendar + customer drivers	13.97%	7
Seasonal naive (repeat same week last year)	not enough history	0



Month	Forecast	Range
2026-04	8,527,531	7,466,764 – 9,588,297
2026-05	7,277,674	5,809,206 – 8,746,142
2026-06	7,682,696	5,760,866 – 9,604,525

9. AI Business Insights

AI insights are unavailable right now (gemini: Gemini rate limit exceeded — too many requests. Try again shortly.; ollama fallback: The AI service is currently unavailable.). The rest of this report contains verified analytical results only.

10. Limitations

- Model predictions are estimates based on historical patterns and may not hold if underlying business conditions change. They should inform, not replace, domain expertise.

11. Appendix

Dataset statistics

Statistic	Value
Rows	69,785
Columns	12